

100% Excellent!

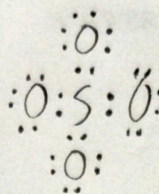
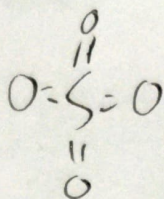
Name: _____

These questions assess your understanding of the material from Chapters 1, 2 of Openstax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Provide the chemical formula for the molecular compound sulfur tetroxide.

ANSWER: SO₄

monosulfur tetroxide
1 4



2. A pharmacist measures the mass of a powdered drug. The mass is 8251 mg. What is the mass in grams? Report your answer in scientific notation.

ANSWER: 8.251 x 10⁰ g

$$8251 \times 10^{-3} = 8.251 \text{ g}$$

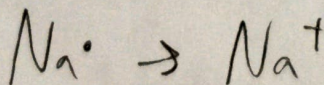
3. Rachel needs to know the mass of a sample of (H₃O)₂O to 4 significant figures. She puts the sample on a digital scale. This is what the scale shows: 50.78 g. If this measurement is precise enough for Rachel, round it to 4 significant figures and report your answer in scientific notation. Remember to include units in your answer.

ANSWER: 5.078 x 10¹ g

$$\begin{aligned} 50.78 &= 4 \text{ sig figs} \\ 50.78 &= 5.078 \times 10^1 \end{aligned}$$

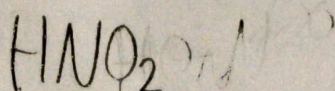
4. Provide the chemical symbol of the most commonly formed ion of sodium and indicate whether it is a cation or an anion. Make sure you include its charge.

ANSWER: Na⁺, cation



5. Write the chemical formula for nitrous acid.

ANSWER: HNO₂



6. Classify Li_2CO_3 as either an ionic or a molecular compound. Explain why you chose your answer.

ANSWER: Ionic

Ionic, dissociates into Li^+ and CO_3^{2-} ions.

7. Efe looks up the density of a material and finds it listed as $2.0 \frac{\text{g}}{\text{cm}^3}$. Convert this density to $\frac{\text{kg}}{\text{m}^3}$. Show your work. Ensure your answer has the correct number of significant figures.

ANSWER: $2.0 \times 10^3 \text{ kg/m}^3$

$$\frac{2.0 \text{ g}}{\text{cm}^3} \times \frac{1 \text{ kg}}{1000 \text{ g}} \times \frac{1,000,000 \text{ cm}^3}{1 \text{ m}^3}$$

$$\frac{2,000,000}{1,000} = \frac{2,000 \text{ kg}}{\text{m}^3}$$

8. As part of his duties as an analytical chemist, Jacob analyzes a sample of an unknown pure substance X and finds that it contains 6.0 g of element E_1 and 7.8 g of element E_2 , and no other elements. He then analyzes three other substances that also contain only elements E_1 and E_2 and records the results in a table in his lab notebook, reproduced below. Which of the other three substances is also a pure substance? Explain why you chose your answer.

sample	mass of E_1	mass of E_2
X	6.0 g	7.8 g
A	25.9 g	15.3 g
B	15.4 g	13.5 g
C	12.3 g	5.6 g

ANSWER: Mixture

$$\frac{12.3}{5.6} \neq \frac{15.4}{13.5}$$

different mass ratios between samples

9. A geologist measures the temperature of a hot spring. The temperature of the spring is 96.3°C . Convert this temperature to kelvin. Report your answer with the correct number of significant figures.

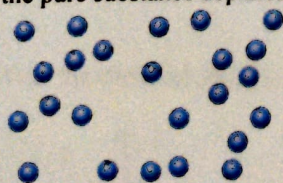
ANSWER: 369.5°K

$$96.3 + 273.15 = 369.45$$

note: "45" rounds to "4" in chemistry (banker's rounding)

1 trailing unit

10. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture? Explain why you chose your answer.



Element, as there is no bonding between atoms, and they cannot be broken down into simpler parts.

ANSWER: Element

(Aside from nuclear reactions)

However, it is possible it is a mixture of noble gasses

+5

11. What multiplicative factor does the metric prefix 'kilo' (symbol: k) represent?

ANSWER:

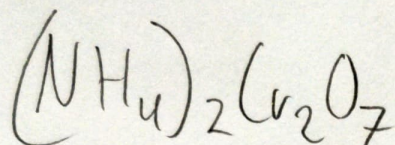
10^3 or 1000

kilo = 10^3

12. Provide the name and empirical formula of the ionic compound that is formed by the NH_4^+ cation and the $\text{Cr}_2\text{O}_7^{2-}$ anion.

ANSWER:

Ammonium Dichromate, $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$



13. As part of her duties as an analytical chemist, Patricia measures the amounts of elements E_1 and E_2 found in four samples collected from an unknown substance X. She then records these masses in a table in her lab notebook, reproduced below. It is known that X contains no elements other than E_1 and E_2 . Is X a pure substance or a mixture? If X is a pure substance, calculate the mass of element E_1 that Patricia would find in a new 10.0-g sample of X. Round your answer to two significant figures. Explain why you chose your answer.

sample	mass of E_1	mass of E_2
1	12.0 g	9.0 g
2	10.5 g	8.8 g
3	11.8 g	5.9 g
4	7.5 g	4.2 g

ANSWER:

Mixture

$$\frac{12.0}{9.0} \neq \frac{11.8}{5.9}$$

Unequal mass ratios shows different compounds. 1932

14. Classify the following as a homogeneous or heterogeneous mixture: a solution of salt dissolved in water. You must explain your answer and use definitions to support your conclusion.

ANSWER:

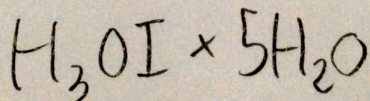
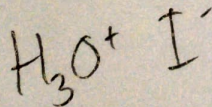
Homogeneous

Homogeneous, on a microscopic scale there is uniform composition.

15. Write the chemical formula of hydronium iodide pentahydrate.

ANSWER:

$\text{H}_3\text{OI} \cdot 5\text{H}_2\text{O}$



+5

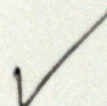
16. Find the difference in the following measurements, ensuring that your answer has the correct number of significant figures.

$$74.7 \text{ g} - 2.408 \text{ g} = 72.292$$

1 unit

ANSWER:

72.3 g



17. Provide the name of the element that the element symbol U represents.

ANSWER:

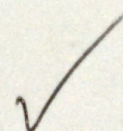
Uranium



18. Identify the state of matter in which water exists at 25 °C and 1 atm.

ANSWER:

liquid

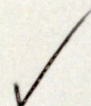


25°C
Solid 0°C < liquid < 100°C gas
at 1 atm

19. Linda needs to determine the mass of a sample of motor oil. The density of the sample is $2.0 \frac{\text{g}}{\text{mL}}$ and its volume is 14.8 mL. What is the mass of the sample? Ensure your answer has the correct number of significant figures.

ANSWER:

30 g



$$2.0 \frac{\text{g}}{\text{mL}} \cdot 14.8 \text{ mL} = 29.6 \text{ g}$$

2 sig figs rounds to 30 g

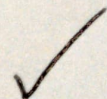
20. Multiply the following measurements, ensuring that your answer has the correct number of significant figures.

$$30 \frac{\text{mol}}{\text{L}} \times 91.3 \text{ L}$$

1 sig fig

ANSWER:

3000 mol



2739

1. SO_4
2. $8.251 \times 10^0 \text{ g}$
3. $5.078 \times 10^1 \text{ g}$
4. Sodium most commonly forms the cation Na^+ .
5. HNO_2
6. Li_2CO_3 is an ionic compound.
7. $2.0 \times 10^3 \frac{\text{kg}}{\text{m}^3}$
8. None of the substances are pure substances.
9. 369.4 K
10. Element
11. 10^3
12. Ammonium dichromate: $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$
13. X is a mixture.
14. Homogeneous mixture $\because$ it is a combination of two or more substances with a uniform composition and appearance.
15. Hydronium iodide pentahydrate: $\text{H}_3\text{OI} \cdot 5\text{H}_2\text{O}$
16. 72.3 g
17. Uranium
18. Liquid
19. 30. g
20. $3 \times 10^3 \text{ mol}$